

Freezing the Web: A Study of ReDoS Vulnerabilities in JavaScript-based Web Servers

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Freezing the Web: A Study of ReDoS Vulnerabilities in JavaScript-based Web Servers

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Regular expression denial of service (ReDoS) is a class of algorithmic complexity attacks where matching a regular expression against an attacker-provided input takes unexpectedly long. The single-threaded execution model of JavaScript makes JavaScript-based web servers particularly susceptible to ReDoS attacks. Despite this risk and the increasing popularity of the server-side Node.js platform, there is currently little reported knowledge about the severity of the ReDoS problem in practice. This paper presents a large-scale study of ReDoS vulnerabilities in real-world web sites. Underlying our study is a novel methodology for analyzing the exploitability of deployed servers. The basic idea is to search for previously unknown vulnerabilities in popular libraries, hypothesize how these libraries may be used by servers, and to then craft targeted exploits. In the course of the study, we identify 25 previously unknown vulnerabilities in popular modules and test 2,846 of the most popular websites against them. We find that 339 of these web sites suffer from at least one ReDoS vulnerability. Since a single request can block a vulnerable site for several seconds, and sometimes even much longer, ReDoS poses a serious threat to the availability of these sites. Our results are a call-to-arms for developing techniques to detect and mitigate ReDoS vulnerabilities in JavaScript.

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